

W2 PRACTICE

Native HTTP and Manual Routing

 *At the end of this practice, you can*

- ✓ **Create** and run a native Node.js HTTP server
- ✓ **Manually implement** route handling using conditionals.
- ✓ Serve **static files** using fs.
- ✓ Parse form data from POST requests.
- ✓ Debug and enhance server code using console outputs.

 *Get ready before this practice!*

- ✓ **Read** the following documents to understand Nodejs built-in HTTP module:
<https://nodejs.org/api/http.html>
- ✓ **Read** the following documents to understand Anatomy of an HTTP Transaction:
<https://nodejs.org/en/learn/modules/anatomy-of-an-http-transaction>

 *How to submit this practice?*

- ✓ Once finished, push your **code to GITHUB**
- ✓ Join the **URL of your GITHUB** repository on LMS



EXERCISE 1 – ANALYZE

Goal

- ✓ Identify and fix the bug.
- ✓ Understand the request-response cycle in Node.js using the `http` module.
- ✓ Explain the role of `res.write()` and `res.end()` in sending data back to the client.



For this exercise, you are provided with a minimal `server.js` file. Read and run the code and observe how it behaves.

```
// server.js
const http = require('http');

const server = http.createServer((req, res) => {
  res.write('Hello, World!');
  return res.endd();
});

server.listen(3000, () => {
  console.log('Server running on http://localhost:3000');
});
```

Q1 – What error message do you see in the terminal when you access `http://localhost:3000`? What line of code causes it?

- `Res.endd()` is not a function.
- Line 5 cause it change it to `res.end()`;

Q2 – What is the purpose of `res.write()` and how is it different from `res.end()`?

- Write or send content into server
- End use to close connection between user and server

Q3 – What do you think will happen if `res.end()` is not called at all?!

- Can lead to memory leak

Q4 – Why do we use `http.createServer()` instead of just calling a function directly?

- Function can be run once, while `http.createServer()` can handle many request.

Q5 – How can the server be made more resilient to such errors during development?

- by testing many test case
- using third party like cloud flare or amazon web service
- monitoring the system and do have backups.

EXERCISE 2 – MANIPULATE

Goal

- ✓ Practice using `req.url` and `req.method`.
- ✓ Understand how manual routing mimics what frameworks (like Express) automate.
- ✓ Serve both plain text and raw HTML manually.

☛ For this exercise you will start with a **START CODE (EX-2)**

TASK 1 - Update the code above to add custom responses for these routes:

ROUTE	HTTP METHOD	RESPONSE
/about	GET	About us: at CADT, we love node.js!
/contact-us	GET	You can reach us vai email...
/products	GET	Buy one get one...
/projects	GET	Here are our awesome projects

Use [VS Code's Thunder Client](#) (or other tools (POSTMAN, INSOMIA) of your choice or curl on your terminal to make request.

Example output

```
curl http://localhost:3000/about ----->
About us: at CADT, we love node.js!
```

```
curl http://localhost:3000/contact-us -----
-> You can reach us vai email...
```

TASK 2 – As we can see the complexitiy grow as we add more routes. Use `switch` statement to arrange the code into more organized structure.

? Reflective Questions

1. What happens when you visit a URL that doesn't match any of the three defined?
 - It will lead user to 404 page or else condition.
2. Why do we check both the `req.url` and `req.method`?

- Because server need to check which url is begin used and used by which method(POST, GET, PATCH, DELETE)
3. What MIME type (`Content-Type`) do you set when returning HTML instead of plain text?
- `Content-Type: text/html`: to inform the server that we going to put html to the server.
4. How might this routing logic become harder to manage as routes grow?
- As routes grow it become hard to control because
 - o To many different method and url
 - o Hard to manage and maintaine
 - o Slow performance
5. What benefits might a framework offer to simplify this logic?
- Have clear folder structure: make it easier to manage, maintenance and debug.
 - Make it more secure

EXERCISE 3 – CREATE

Goal

- ✓ Practice handling `POST` requests.
- ✓ Parse URL-encoded form data manually.
- ✓ Write and append to local files using Node.js' `fs` module.
- ✓ Handle async operations and errors gracefully.

🔗 For this exercise you will start with a **START CODE EX-3**

TASK 1 - Extend your Node.js HTTP server to handle a **POST request** submitted from the contact form. When a user submits their name, the server should:

1. **Capture the form data** (from the request body).
2. **Log it to the console.**
3. **Write it to a local file** named `submissions.txt`.

Testing, go to `/contact` on browser and test

Requirements

- Handle `POST /contact` requests.
- Parse raw `application/x-www-form-urlencoded` data from the request body.
- Write the name to a new line in `submissions.txt`.
- Send a success response to the client (HTML or plain text).

- `fs.appendFile`: is add data into the file and keep existing data. If that file have some data in it, it is not overwrite over the old data, while `fs.writeFile`, it will overwrite over the old data.

5. How could this be improved or made more secure?

- Handle many request at the same time
- Handle error that can happen
- Validate user input

Bonus Challenge (Optional)

- Validate that the `name` field is not empty before saving.
- Send back a small confirmation HTML page instead of plain text.
- Try saving submissions in JSON format instead of plain text.